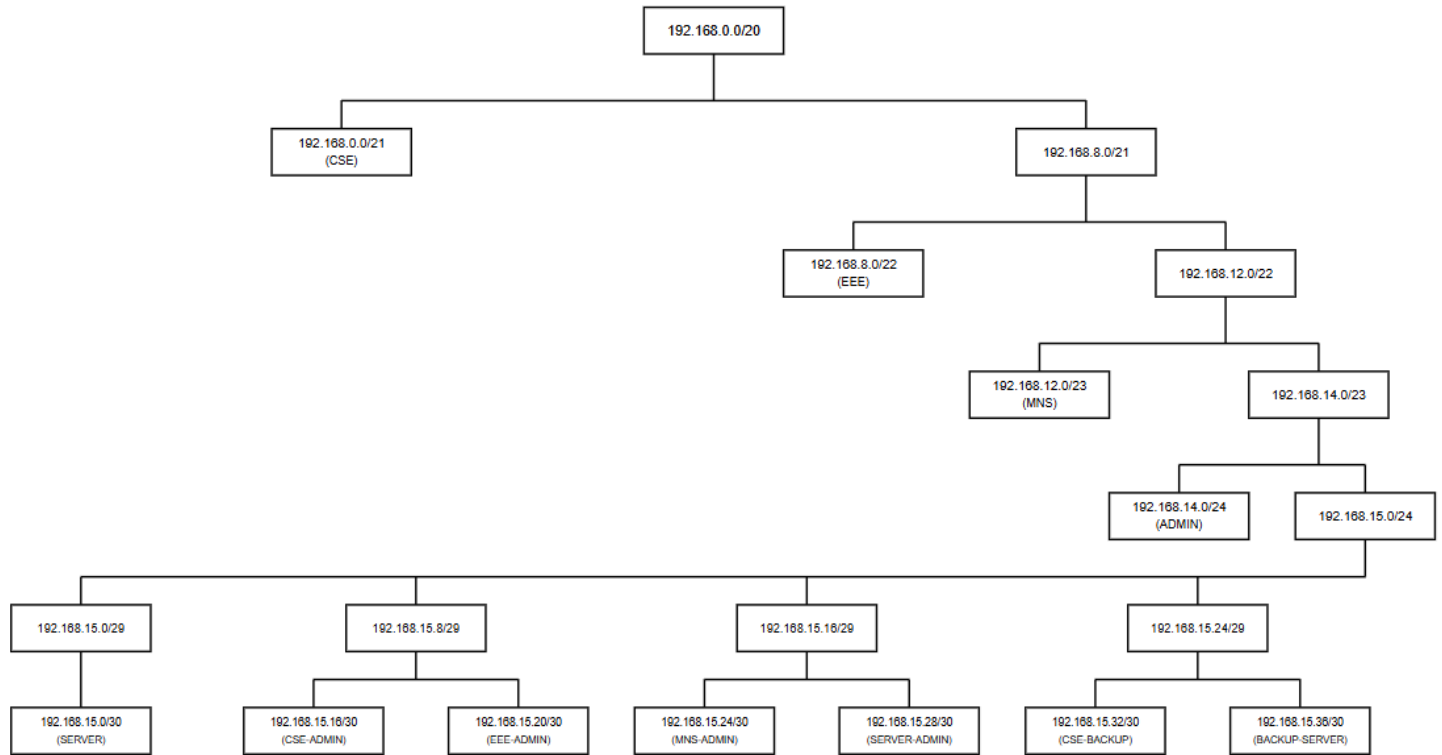


VLSM Tree



IP Addressing table

LAN Networks

Building	Network Address	Subnet Mask	Usable Range	Device	IP Address	Gateway	Config
CSE	192.168.0.0/21	255.255.248.0	192.168.0.1 – 192.168.7.254	Router (Gig0/0)	192.168.0.1	–	Static
				PC1 & PC2	DHCP	192.168.0.1	DHCP
				SuperComputer	192.168.7.254	192.168.0.1	Static
EEE	192.168.8.0/22	255.255.252.0	192.168.8.1 – 192.168.11.254	Router (Gig0/0)	192.168.8.1	–	Static
				PC1 & PC2	DHCP	192.168.8.1	DHCP
				SuperComputer	192.168.11.254	192.168.8.1	Static
MNS	192.168.12.0/23	255.255.254.0	192.168.12.1 – 192.168.13.254	Router (Gig0/0)	192.168.12.1	–	Static
				PC1 & PC2	DHCP	192.168.12.1	DHCP
				SuperComputer	192.168.13.254	192.168.12.1	Static
Admin	192.168.14.0/24	255.255.255.0	192.168.14.1 – 192.168.14.254	Router (Gig0/0)	192.168.14.1	–	Static
				PC1 & PC2	DHCP	192.168.14.1	DHCP
				SuperComputer	192.168.14.254	192.168.14.1	Static
Server	192.168.15.0/29	255.255.255.248	192.168.15.1 – 192.168.15.6	Router (Gig0/0)	192.168.15.1	–	Static
				Web Server	192.168.15.2	192.168.15.1	Static
				DNS Server	192.168.15.3	192.168.15.1	Static
				Email Server	192.168.15.4	192.168.15.1	Static
				DHCP Server	192.168.15.5	192.168.15.1	Static

Point-to-Point Links

Link Name	Network Address	Subnet Mask	Router Interface 2	Router Interface 1
CSE-Admin	192.168.15.16/30	255.255.255.252	Admin (Ser0/1/0)	CSE (Ser0/1/0)
EEE-Admin	192.168.15.20/30	255.255.255.252	Admin (Ser0/1/1)	EEE (Ser0/1/0)
MNS-Admin	192.168.15.24/30	255.255.255.252	Admin (Ser0/0/0)	MNS (Ser0/1/0)
Server-Admin	192.168.15.28/30	255.255.255.252	Server (Ser0/1/0)	Admin (Ser0/0/1)
CSE-Backup	192.168.15.32/30	255.255.255.252	Backup (Ser0/1/0)	CSE (Ser0/1/1)
Backup-Server	192.168.15.36/30	255.255.255.252	Server (Ser0/1/1)	Backup (Ser0/1/1)

CSE ROUTER - CONFIGURATION

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname CSE_Router
```

```
CSE_Router(config)#interface GigabitEthernet0/0
```

```
CSE_Router(config-if)#ip address 192.168.0.1 255.255.248.0
```

```
CSE_Router(config-if)#ip helper-address 192.168.15.5
```

```
CSE_Router(config-if)#no shutdown
```

```
CSE_Router(config-if)#exit
```

```
CSE_Router(config)#interface Serial0/1/0
```

```
CSE_Router(config-if)#ip address 192.168.15.17 255.255.255.252
```

```
CSE_Router(config-if)#no shutdown
```

```
CSE_Router(config-if)#exit
```

```
CSE_Router(config)#interface Serial0/1/1
```

```
CSE_Router(config-if)#ip address 192.168.15.33 255.255.255.252
```

```
CSE_Router(config-if)#clock rate 64000
```

```
CSE_Router(config-if)#no shutdown
```

```
CSE_Router(config-if)#exit
```

```
CSE_Router(config)#router rip
```

```
CSE_Router(config-router)#version 2
```

```
CSE_Router(config-router)#passive-interface Serial0/1/1
```

```
CSE_Router(config-router)#network 192.168.0.0
```

```
CSE_Router(config-router)#network 192.168.15.0
```

```
CSE_Router(config-router)#no auto-summary
```

```
CSE_Router(config-router)#exit
```

```
CSE_Router(config)#ip route 192.168.15.0 255.255.255.248 192.168.15.34 130
```

```
CSE_Router(config)#end
```

```
CSE_Router#write memory
```

EEE ROUTER - CONFIGURATION

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname EEE_Router
```

```
EEE_Router(config)#interface GigabitEthernet0/0
```

```
EEE_Router(config-if)#ip address 192.168.8.1 255.255.252.0
```

```
EEE_Router(config-if)#ip helper-address 192.168.15.5
```

```
EEE_Router(config-if)#no shutdown
```

```
EEE_Router(config-if)#exit
```

```
EEE_Router(config)#interface Serial0/1/0
```

```
EEE_Router(config-if)#ip address 192.168.15.21 255.255.255.252
```

```
EEE_Router(config-if)#clock rate 64000
```

```
EEE_Router(config-if)#no shutdown
```

```
EEE_Router(config-if)#exit
```

```
EEE_Router(config)#router rip
```

```
EEE_Router(config-router)#version 2
```

```
EEE_Router(config-router)#network 192.168.8.0
```

```
EEE_Router(config-router)#network 192.168.15.0
```

```
EEE_Router(config-router)#no auto-summary
```

```
EEE_Router(config-router)#exit
```

```
EEE_Router(config)#end
```

```
EEE_Router#write memory
```

MNS ROUTER - CONFIGURATION

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname MNS_Router
```

```
MNS_Router(config)#interface GigabitEthernet0/0
```

```
MNS_Router(config-if)#ip address 192.168.12.1 255.255.254.0
```

```
MNS_Router(config-if)#ip helper-address 192.168.15.5
```

```
MNS_Router(config-if)#no shutdown
```

```
MNS_Router(config-if)#exit
```

```
MNS_Router(config)#interface Serial0/1/0
```

```
MNS_Router(config-if)#ip address 192.168.15.25 255.255.255.252
```

```
MNS_Router(config-if)#no shutdown
```

```
MNS_Router(config-if)#exit
```

```
MNS_Router(config)#router rip
```

```
MNS_Router(config-router)#version 2
```

```
MNS_Router(config-router)#network 192.168.12.0
```

```
MNS_Router(config-router)#network 192.168.15.0
```

```
MNS_Router(config-router)#no auto-summary
```

```
MNS_Router(config-router)#exit
```

```
MNS_Router(config)#end
```

```
MNS_Router#write memory
```

ADMIN ROUTER - CONFIGURATION

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname Admin_Router
```

```
Admin_Router(config)#interface GigabitEthernet0/0
```

```
Admin_Router(config-if)#ip address 192.168.14.1 255.255.255.0
```

```
Admin_Router(config-if)#ip helper-address 192.168.15.5
```

```
Admin_Router(config-if)#no shutdown
```

```
Admin_Router(config-if)#exit
```

```
Admin_Router(config)#interface Serial0/1/0
```

```
Admin_Router(config-if)#ip address 192.168.15.18 255.255.255.252
```

```
Admin_Router(config-if)#clock rate 64000
```

```
Admin_Router(config-if)#no shutdown
```

```
Admin_Router(config-if)#exit
```

```
Admin_Router(config)#interface Serial0/1/1
```

```
Admin_Router(config-if)#ip address 192.168.15.22 255.255.255.252
```

```
Admin_Router(config-if)#clock rate 64000
```

```
Admin_Router(config-if)#no shutdown
```

```
Admin_Router(config-if)#exit
```

```
Admin_Router(config)#interface Serial0/2/0
```

```
Admin_Router(config-if)#ip address 192.168.15.25 255.255.255.252
```

```
Admin_Router(config-if)#clock rate 64000
```

```
Admin_Router(config-if)#no shutdown
```

```
Admin_Router(config-if)#exit
```

```
Admin_Router(config)#interface Serial0/2/1
```

```
Admin_Router(config-if)#ip address 192.168.15.30 255.255.255.252
```

```
Admin_Router(config-if)#clock rate 64000
```

```
Admin_Router(config-if)#no shutdown
```

```
Admin_Router(config-if)#exit
```

```
Admin_Router(config)#router rip
```

```
Admin_Router(config-router)#version 2
```

```
Admin_Router(config-router)#network 192.168.0.0
```

```
Admin_Router(config-router)#network 192.168.8.0
```

```
Admin_Router(config-router)#network 192.168.12.0
```

```
Admin_Router(config-router)#network 192.168.14.0
```

```
Admin_Router(config-router)#network 192.168.15.0
```

```
Admin_Router(config-router)#no auto-summary
```

```
Admin_Router(config-router)#exit
```

```
Admin_Router(config)#ip route 192.168.15.0 255.255.255.248 192.168.15.30
```

```
Admin_Router(config)#end
```

```
Admin_Router#write memory
```

SERVER ROUTER - CONFIGURATION

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname Server_Router
```

```
Server_Router(config)#interface GigabitEthernet0/0
```

```
Server_Router(config-if)#ip address 192.168.15.1 255.255.255.248
```

```
Server_Router(config-if)#no shutdown
```

```
Server_Router(config-if)#exit
```

```
Server_Router(config)#interface Serial0/1/0
```

```
Server_Router(config-if)#ip address 192.168.15.29 255.255.255.252
```

```
Server_Router(config-if)#no shutdown
```

```
Server_Router(config-if)#exit
```

```
Server_Router(config)#interface Serial0/1/1
```

```
Server_Router(config-if)#ip address 192.168.15.37 255.255.255.252
```

```
Server_Router(config-if)#clock rate 64000
```

```
Server_Router(config-if)#no shutdown
```

```
Server_Router(config-if)#exit
```

```
Server_Router(config)#ip route 192.168.0.0 255.255.248.0 192.168.15.30
```

```
Server_Router(config)#ip route 192.168.8.0 255.255.252.0 192.168.15.30
```

```
Server_Router(config)#ip route 192.168.12.0 255.255.254.0 192.168.15.30
```

```
Server_Router(config)#ip route 192.168.14.0 255.255.255.0 192.168.15.30
```

```
Server_Router(config)#ip route 192.168.15.32 255.255.255.252 192.168.15.38
```

```
Server_Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.15.30
```

```
Server_Router(config)#ip route 192.168.0.0 255.255.248.0 192.168.15.38 1
```

```
Server_Router(config)#end
```

```
Server_Router#write memory
```


BACKUP ROUTER - CONFIGURATION

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname Backup_Router
```

```
Backup_Router(config)#interface Serial0/1/0
```

```
Backup_Router(config-if)#ip address 192.168.15.34 255.255.255.252
```

```
Backup_Router(config-if)#clock rate 64000
```

```
Backup_Router(config-if)#no shutdown
```

```
Backup_Router(config-if)#exit
```

```
Backup_Router(config)#interface Serial0/1/1
```

```
Backup_Router(config-if)#ip address 192.168.15.38 255.255.255.252
```

```
Backup_Router(config-if)#no shutdown
```

```
Backup_Router(config-if)#exit
```

```
Backup_Router(config)#ip route 192.168.0.0 255.255.248.0 192.168.15.33
```

```
Backup_Router(config)#ip route 192.168.8.0 255.255.252.0 192.168.15.33
```

```
Backup_Router(config)#ip route 192.168.12.0 255.255.254.0 192.168.15.33
```

```
Backup_Router(config)#ip route 192.168.14.0 255.255.255.0 192.168.15.33
```

```
Backup_Router(config)#ip route 192.168.15.0 255.255.255.248 192.168.15.37
```

```
Backup_Router(config)#router rip
```

```
Backup_Router(config-router)#version 2
```

```
Backup_Router(config-router)#network 192.168.15.0
```

```
Backup_Router(config-router)#no auto-summary
```

```
Backup_Router(config-router)#exit
```

```
Backup_Router(config)#router rip
```

```
Backup_Router(config-router)#passive-interface Serial0/1/0
```

```
Backup_Router(config-router)#exit
```

```
Backup_Router(config)#end
```

```
Backup_Router#write memory
```